

OPTRONIC SENSORS: Chemical, Biological And Explosive Sensors



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The need and relevance of efficient and reliable technologies for stand-off detection and identification of chemical, biological and explosive agents have become increasingly important in present-day scenario. Chemical and biological weapons are relatively inexpensive to produce and can unleash a devastating effect as terrorist weapons.

The international community has already witnessed devastating effects these weapons can cause on more than one occasion. Millions have been affected by the destruction caused by these weapons of mass destruction in the past. Sarin attack by Aum Shinrikyo cult in 1995 in a Tokyo subway, hydrogen cyanide and mustard gas attack by Iraq in Anfal campaign against Kurds most notably in Halabja massacre in 1988 (Fig. 1) and international dispersal of anthrax spores against the US in 2001 are some examples of chemical and biological weapon attacks.

Use of improvised explosive devices (IEDs) by terrorists and anti-national elements killing soldiers and civilians alike has become a routine affair. The incidents have brought to the fore the need for stand-off detection and identification of chemical, biological and explosive agents. A number of laser-based methodologies/techniques are either in use or being explored for the purpose.

Detection of chemical warfare agents

Differential absorption lidar. Nerve agents (such as organophosphate compounds) and blister agents (such as mustard compounds), because of their acute toxicity, rapidity of action, undetectability by human senses and economic viability, are two potentially harmful classes of chemical warfare agents.

Nerve agents are organophosphorus esters, which initially stimulate and then paralyse certain nerve transmissions by interfering with cholinesterase enzymes. Common chemical warfare agents belonging to this category are tabun, sarin, soman, Vx, dichlorvos and malathion. Fig. 2 shows typical effects of a chemical warfare agent attack.

Lidar-based detection is the most practically realisable technique for stand-off

detection of chemical and biological warfare agents. Commonly exploited physical phenomena forming the basis of chemical and biological agent detection include elastic backscattering, laser-induced fluorescence (LIF) and differential absorption.

These are superior to point detection systems such as infrared (IR) spectroscopy and Raman spectroscopy. This is due to their capability to range and discriminate chemical and biological molecules in real time.

Ultraviolet (UV) LIF can be used for the detection of biological molecules since most of these fluoresce when excited by UV radiation.

Differential absorption lidar (DIAL) is the most frequently used technique for the detection of chemical warfare agents along with detection of toxic gases and pollutants. It uses two wavelengths: one corresponds to



Fig. 1: Halabja massacre



Fig. 2: Effects of chemical weapon attack

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